
Cálculo para a Engenharia

Formulário

ENGTI 2022-2023

Derivadas

$$(u^n)' = nu^{n-1}u'$$

$$(\ln u)' = \frac{u'}{u}$$

$$(e^u)' = u'e^u$$

$$(a^u)' = u'a^u \ln a$$

$$(\cos u)' = -u' \sin u$$

$$(\sin u)' = u' \cos u$$

$$(\arcsin u)' = \frac{u'}{\sqrt{1-u^2}}$$

$$(\arccos u)' = -\frac{u'}{\sqrt{1-u^2}}$$

$$(\arctan u)' = \frac{u'}{1+u^2}$$

Primitivas

$$\int u' u^n dx = \frac{u^{n+1}}{n+1} + c$$

$$\int \frac{u'}{u} dx = \ln |u| + c$$

$$\int u' e^u dx = e^u + c$$

$$\int u' a^u dx = \frac{a^u}{\ln a} + c$$

$$\int u' \cos u dx = \sin u + c$$

$$\int u' \sin u dx = -\cos u + c$$

$$\int \frac{u'}{\sqrt{1-u^2}} dx = \arcsin u + c$$

$$\int -\frac{u'}{\sqrt{1-u^2}} dx = \arccos u + c$$

$$\int \frac{u'}{1+u^2} dx = \arctan u + c$$

Fórmulas Trigonométricas

$$\sin^2 x + \cos^2 x = 1$$

$$\tan^2 x + 1 = \frac{1}{\cos^2 x}$$

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$